Campus Pulse

A live campus food-and-seat decision tool. It turns a camera view of a canteen into free-seat counts, queue estimates and one clear answer: where to go right now.

5	192	1–8	3
venues live	seats modelled	group sizes	preferences, any mix

The problem

At lunch, students walk between canteens without knowing which one has a free table, a long queue or a quiet corner. Existing menus show food, not capacity. Campus Pulse answers the only question that matters before you start walking: **where do I actually go?**

What it does

The user picks a group size (1–8) and any combination of **closest**, **quietest** and **no queue**. The app names one venue in large type and shows live free seats, queue wait, walking time, a seat map, ghost seats and the rest of campus. A built-in assistant answers plain-language questions — including why it chose that place — from the same live data.

How the live data is produced

With a camera

A worker reads frames at about 2 fps, runs a pre-trained YOLO11 model, matches people and left-behind items to marked seat zones, and stores only seat states and timers. The frame is discarded immediately. No image, face or person box is ever sent to the browser.

Without a camera

A campus simulator drives every venue from realistic daily curves, service speeds and abandonment behaviour, so all screens work on a laptop with nothing plugged in. Camera-backed seats are labelled separately in the snapshot.

Design principles

- **One answer, not a directory.** The first screen after the questions is a single venue, not a list.
- **The type is the product.** Huge rounded Nunito wordmark and venue names; everything secondary stays small.
- **Nothing pops up uninvited.** Alerts collect on an Alerts screen; the user opens them.
- **No reservations or queue tickets.** The current app informs and recommends; it does not claim seats or join queues.
- **Privacy is structural.** The browser API has no route that can return a frame.

Technology

LAYER	STACK	RESPONSIBILITY
Frontend	Plain HTML, CSS, JavaScript; bundled Nunito woff2	Landing, recommendation, seat map, alerts, staff view, chat; no build step.
Server	Python standard library HTTP server + Server-Sent Events	Static files, JSON snapshot each second, chat endpoint, config, ticket-free API.
Engine	Python: clock, simulators, trackers, forecasts	One source of truth for seat states, queues, alerts and recommendations.
Vision	OpenCV + Ultralytics YOLO11 (COCO), PyAV for streams	People, chairs and left-behind items to seat zones to states.

Campus Pulse — live seats, queues and calm.

Student experience

Every screen a student sees, in the order they see it.

01 · Landing wordmark BRAND

A giant “Campus Pulse” in the same rounded font, with a soft 3D shadow. The letters rise into place once on load; the animation does not replay on data updates. The only other element is the hamburger menu in the top right.

02 · Group size, 1–8 QUESTION

Eight large circular buttons select the number of people. The choice feeds the recommendation, the table-fit search and every answer from the assistant.

03 · Multi-select preferences QUESTION

Closest to me, Quietest and No queue can be selected independently or together. All selected preferences are scored in one ranking function — selecting three does not silently behave like selecting only the last one.

04 · Actions QUESTION

“Show me where to go” requires a group size and opens the recommendation. “Just browsing” opens the same screen with the rest of campus already unfolded.

05 · One recommendation ANSWER

The winning venue is named in display type. A small line above states the group size and the exact preference combination that produced it. Below, one fact line combines free seats, queue state and walking time.

06 · Three live metrics ANSWER

Seats free, queue wait and walk time are updated in place every second, without rebuilding the page or replaying animations. The queue wait uses people in line × average service seconds.

07 · Why this pick ANSWER

“Ask why” opens the assistant and asks about this venue with its ID. The reply explains the seat count, walk, queue, which preferences matched, a genuinely later quiet window and the next two alternatives.

08 · Today’s context ANSWER

A one-line history (“quietest around …” only when that time is actually later and quieter, or “steady for the next two hours”) plus a sparkline of free seats over the last few minutes.

09 · More seats on campus ANSWER

One arrow expands the rest: campus totals, every food venue sorted by the same preferences, study spaces, ghost seats, and shortcuts to the map and alerts. The arrow can hide it again.

10 · Venue detail VENUE

Capacity ring, live queue and seats-taken stats, a two-hour forecast chart with a draggable time slider, a group-seat finder with “Hold these seats” sharing code, the full seat map and ghost-seat timers. No seat-claiming or queue-joining action exists.

11 · Seat map VENUE

Seats are coloured by state: free, someone sitting, bag with owner nearby, abandoned, and camera-observed. Tables are drawn from the floor plan. Group seats are circled. Tapping a seat reports its state; no camera image is involved.

12 · Ghost seats VENUE

When an item is seen with no person for longer than the venue’s threshold (15–25 minutes; 20 minutes for most venues), the seat changes from held to probably free. A live timer shows how long the item has been alone. This is information only — the user cannot claim it.

Everything else

The screens behind the hamburger, the assistant, and the operational view.

13 · Hamburger menu NAVIGATION

Three lines in the top right open Home, Map, Alerts, Staff view, Demo controls and Ask Campus Pulse. The Student/Staff segmented switch sits at the top of this menu. There is no other top navigation, clock or status chrome on the page.

14 · Campus map MAP

A schematic campus with live pins showing each venue's free-seat count and a colour for crowd level (quiet, filling, busy, packed). A nearest-first list below gives walk time, distance, opening hours and free seats.

15 · Alerts ALERTS

Computed alerts appear in one place: go-now advice, a venue about to fill up, and a seat whose item has been abandoned. The user chooses whether rush or ghost alerts are enabled. No browser notification fires on its own.

16 · Student / Staff switch ROLES

Switches the same live snapshot between the student recommendation and the canteen team's operational view. The data is identical; only the explanation and metrics change.

17 · Staff view STAFF

Meals served today (estimated), queueing minutes avoided, peak window, a peak-hours bar chart per venue, busiest and quietest hour, counters to staff at peak, food-waste risk, a week heatmap and a detection-layer panel showing cameras attached, frames processed and seats backed by real detection.

18 · Ask Campus Pulse ASSISTANT

A local rules engine over the live snapshot — no cloud call, no API key. It understands group seats together, named venues, quietest time, queue length, ghost seats, general "where should I eat" and "why this place". It receives group size, selected preferences and an optional focus venue, so its answer agrees with the screen. Replies render bold, line breaks and a typing state; suggested-question chips are clickable and double-sends are blocked.

19 · Demo controls DEMO

Camera status, campus clock speed (1×–120×), day selection, jump-to times (08:00, 13:00, 17:30, 19:30) and scenarios: normal day, lunch rush, ghost seats, near empty. Intended for the operator, not the judge-facing flow.

20 · Live camera mode VISION

Attach a webcam, a phone stream or a recorded clip. The worker reads at 1–3 fps, detects people, chairs and items, and maps boxes to seat zones. Camera-backed seat counts replace the simulator for those seats; the rest stay simulated. An operator preview and an annotated proof recording are optional and never reach the browser.

21 · Privacy by construction PRIVACY

Frames are processed on the device and dropped in the same loop. Snapshots contain only seat states, counts, queue lengths, forecasts and timers. The API has no endpoint that can return an image, a crop or a person box.

The five venues in the current dataset

VENUE	TYPE	SEATS	WALK	OPEN
Nitte Main Canteen	Canteen	48	2 min	07:30–21:30
Annapoorna Veg	Restaurant	32	6 min	11:00–22:30
Juice Junction	Cafe	12	4 min	08:00–21:00
Food Court, Block C	Food court	40	8 min	08:00–20:30
Central Library Reading Hall	Library	60	3 min	08:00–23:00

How it is built

Data flow

Camera frames to YOLO11 detection (people / chairs / items) to seat-zone overlap to **SeatTracker** (free, busy, held, probably free, unknown, timers)
Simulator to the same tracker shape for every seat without a live camera
Engine merges camera + simulator, then computes forecast, crowd level, queue, alerts and staff stats, producing a **snapshot** (JSON) served over **SSE / REST** to the **browser**.
Snapshot to **rules assistant** to chat reply. No frame ever enters this path.

Engine responsibilities

- Owns one campus clock that starts at 08:00 and can run at 1×–120× real time.
- Runs one simulator and one tracker per venue; camera seats are owned by real detections when a camera is delivering frames.
- Computes free seats, crowd level (quiet / filling / busy / packed), queue length, wait minutes, forecasts, quietest window, ghosts, alerts and staff metrics.
- Resyncs the world after a time jump so stale ghost timers and sparklines cannot leak across the jump.
- Exposes one snapshot shape consumed by every screen and by the assistant.

Vision and seat states

STATE	MEANING
free	No person or item detected in the seat zone.
busy	A person intersects the seat zone.
held	An item is present but no person; owner may be nearby, timer just started.
probably_free	An item alone longer than the venue's ghost threshold; the app suggests it is abandoned.
unknown	No camera coverage for that seat; the simulator provides a stand-in.

The detector uses COCO class 0 for people, 56 for chairs, and 24/26/28/63/73 for backpack, handbag, suitcase, laptop and book. Confidence is 0.30 at 640 px by default. The model is pre-trained, not fine-tuned on this campus.

Forecast, alerts and staff metrics

Forecast
A two-hour projection of free seats from the observed daily pattern and venue popularity. The venue screen lets the user drag through it. It is an estimate, not a reservation.

Alerts
"Go now" advice from the top recommendation; "about to fill" when the forecast drops sharply within the next 90 minutes; "ghost seat" when an item has been alone past the threshold.

Staff
Peak occupancy and hour, suggested counters at peak (one per ~14 people per 15 minutes), food-waste risk from the gap between peak and average demand, and estimated meals served.

Queue
People in the queue area × average service seconds per person. The app states this is an estimate and shows it as information only.

Files, start and API

Files

run.py	Starts engine, cameras and HTTP server.
server.py	Static files, snapshot, SSE, chat, squad code, config, health.
cv/engine.py	Clock, merge, forecast, recommendation, alerts, staff, snapshot.
cv/detection.py	YOLO wrapper and operator annotation.
cv/camera.py	Camera workers; the only code that touches pixels.
cv/tracker.py	Seat states, ghost timers, group finder, queue math.
cv/simulator.py	Campus crowd simulation.
cv/insights.py	The assistant, answered from the live snapshot.
web/	HTML, CSS, JS and bundled font; no framework, no build.
data/venues.json	Venues, floor plans, capacities and hours.
data/cameras.json	Camera-to-seat zones.

Start

python run.py — simulator only, opens http://localhost:8000
python run.py --camera 0 --show — laptop webcam, operator preview with boxes
python run.py --camera http://192.168.1.7:8080/video --show — phone as a live camera
python run.py --camera clip.mp4 --record proof.mp4 — recorded clip and proof video
python run.py --port 3000 --no-browser — the port used by the hosted preview
CAMPUSPULSE_MODEL=yolo11s.pt python run.py --camera 0 — larger model, slower and more accurate

Dependencies: **pip install -r requirements.txt** (OpenCV, Ultralytics, NumPy). The web app and API use only the Python standard library.

API

ROUTE	PURPOSE
GET /	The application.
GET /api/snapshot	One full snapshot; optional ?group=N.
GET /api/events	Server-Sent Events; pushes a snapshot every second.
POST /api/chat	message, group, matters, optional venue_id; returns reply + chips.
POST /api/squad	Short share code for a group's seats.
POST /api/config	speed, scenario, day, time.
GET /api/health	Readiness and campus clock.

There is deliberately no seat-claim route and no queue-ticket route in the current build.

What has been verified

- The app runs end to end on the simulator, and on a live public camera feed with **mode: live-cv**, frames processed and counts updating in the interface.
- All screens render in headless Chromium with no console errors; landing, multi-select, recommendation, seat map, menu, chat and staff view were exercised.
- The assistant answers group-size, venue-name, queue, quietest-time, ghost-seat and "why this pick" questions from live data.
- Live-feed counting was measured against hand-counted frames on a public eatery camera.

Accuracy and limits

Measured accuracy, stated honestly

On a wide, dim, crowded open-air eatery at night, the pre-trained model drew real boxes but badly undercounted people. Mean absolute error was **25** people with yolo11n at 640, **16** with yolo11s at 640 and **13** with yolo11s at 960; recall rose from about **19%** to **58%**. This is a wide-shot limitation, not a seat-state failure.

The intended demo is a phone or webcam above one table, where people are large in frame. The app is designed for that distance; it does not claim accurate counting in a crowded hall.

Limits

- One camera sees one area; a large hall needs multiple cameras and zones.
- Occlusion, dim light and shallow angles reduce accuracy; the queue estimate assumes steady service speed.
- Forecasts and the staff heatmap currently come from simulated history until real days of data exist.
- No seat reservations, no queue joining, no payments and no accounts in this build.
- Privacy: no footage, faces or person boxes are stored or exposed; only aggregate seat states leave the process.

Current UI decisions

- No top navigation bar: only the wordmark and the three-line menu button.
- No helper sentences under section headings unless they carry information the user needs.
- The recommendation page updates live values in place, so it does not blink or re-animate every second.
- Ask why explains the recommendation rather than opening an empty chat.